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Roll No

EX-8403 (GS)**B.E. VIII Semester** Examination, June 2020**Grading System (GS)****Advanced Control System****(Elective - IV)****Time : Three Hours****Maximum Marks : 70**

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain the concepts of state variables.
b) Investigate the stability.

$$S^5 + S^4 + 2S^3 + 2S^2 + 3S + 5 = 0$$

2. a) Write a short note on Pole placement by state feedback.

- b) A system has transfer function $\frac{Y(s)}{U(s)} = \frac{9}{s^2 - 9}$. Calculate K so that the control law $= -Kx$, places the closed loop poles at $-3 \pm j3$.

3. a) Distinguish between conventional control theory and modern control theory.
b) Show that the system described below is unstable

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 3 & -2 \end{bmatrix} x$$

verify the result using Lyapunov method.

4. a) Explain the theorem “Region of Attraction” of variable structure control.
b) What are the salient features of variable structure control? and also give its application.

5. Name the various methods for finding stability? Explain any one method in detail.

6. The dynamics of a control system is represented by :

$$\dot{x}_1 = x_2 - x_1(x_1^2 + x_2^2) ; \dot{x}_2 = -x_1 - x_2(x_1^2 + x_2^2)$$

Determine the system stability using Lyapunov method.

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7. Write a short notes on the following

- a) Calculus of variation
- b) Bolza problems

8. Write short notes (any two) :

- a) Euler - Lagrange equations
- b) Pole placement problem
- c) Boundary conditions
